Software Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

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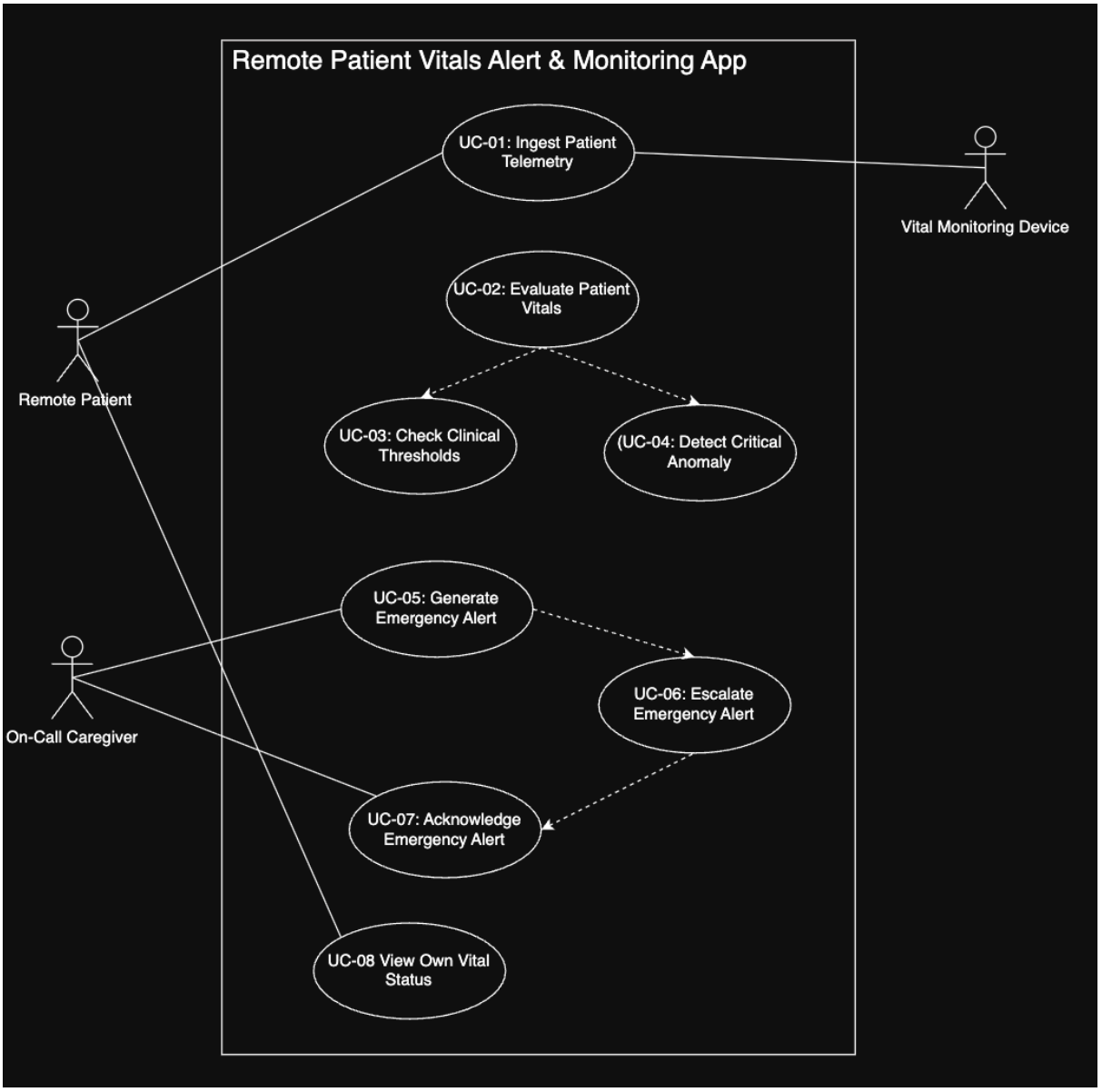
Requirements Table

Req Id	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall continuously evaluate ingested patient vitals against configured clinical thresholds and flag critical anomalies within 2 seconds.	High	**Pass:** A high heart rate above 140 BPM triggers an immediate escalation alert. **Fail:** The metric spike is ignored or delayed by more than 5 seconds.	Enables rapid detection of critical patient conditions.
FR-002	Functional	The system shall continuously ingest SpO_2, Heart Rate, and Blood Pressure telemetry from monitored patients.	High	**Pass:** Valid SpO_2, Heart Rate, and BP readings received from a patient are accepted and available for evaluation. **Fail:** A valid telemetry reading is not received by the monitoring pipeline.	Provides the vital data required for continuous monitoring.
FR-003	Functional	The system shall compare	High	**Pass:** A reading outside its	Allows the system to

		each ingested vital reading with its configured clinical threshold and identify threshold breaches.		configured threshold is marked as a threshold breach. **Fail:** An out-of-range reading is not identified.	identify potentially dangerous changes in patient condition.
FR-004	Functional	The system shall generate an emergency alert when a critical vital anomaly is detected.	High	**Pass:** When a critical anomaly is detected, an emergency alert is created containing the relevant patient and vital information. **Fail:** No alert is generated for a critical anomaly.	Ensures critical conditions are communicated for immediate action.
FR-005	Functional	The system shall escalate emergency alerts according to the configured caregiver matrix.	High	**Pass:** A generated emergency alert is sent to the caregiver assigned according to the configured caregiver matrix. **Fail:** The alert is sent to an incorrect caregiver or is not escalated.	Ensures emergency alerts reach the appropriate on-call caregiver.
NFR-001	Nonfunctional — Performance & Security	The telemetry ingestion gateway shall support at least 500 concurrent continuous telemetry streams with 99.99% uptime.	High	**Pass:** Benchmark testing confirms support for at least 500 concurrent telemetry streams while maintaining 99.99% uptime under the specified test conditions. **Fail:** The gateway cannot meet either target.	Supports reliable continuous monitoring at scale.
NFR-002	Nonfunctional — Performance	The system shall process and evaluate incoming patient telemetry with	High	**Pass:** Telemetry is evaluated within the configured processing time during performance testing. **Fail:**	Low processing latency is necessary for timely emergency

		low latency to support timely anomaly detection and alert escalation.		Telemetry processing consistently exceeds the configured processing limit.	alerts.

Use Case Diagram



Use-Case Flow Specification

Use Case: UC-05 – Generate Emergency Alert

Primary Actor: On-Call Caregiver
Supporting Actor: Vital Monitoring Device
System: Remote Patient Vitals Alert & Monitoring App

1. Preconditions

1. The patient is registered in the system.
2. The patient has a connected vital monitoring device.
3. The system is receiving patient vital data such as SpO_2, Heart Rate and Blood Pressure.
4. Clinical thresholds are already configured for the patient.
5. The caregiver matrix has the required on-call caregiver details.

2. Main Success Scenario

1. The Vital Monitoring Device sends the patient's vital readings to the system.
2. The system receives the SpO_2, Heart Rate and Blood Pressure readings.
3. The system checks whether the received data is valid.
4. The system continuously compares the vital readings with the configured clinical thresholds.
5. The system detects that one of the vital readings has crossed the critical threshold.
6. The system identifies the reading as a critical anomaly.
7. The system generates an emergency alert for the patient.
8. The alert contains the patient information, affected vital sign, measured value and time of the alert.
9. The system checks the caregiver matrix and identifies the current On-Call Caregiver.
10. The system sends the emergency alert to the On-Call Caregiver.
11. The caregiver receives and acknowledges the alert.
12. The use case ends successfully.

3. Alternate Flow – No Response from Caregiver

A1. Caregiver does not respond within the time limit

1. The system sends the emergency alert to the assigned On-Call Caregiver.
2. The configured response time passes without receiving an acknowledgement.
3. The system detects that the caregiver has not responded.

4. The system escalates the alert according to the caregiver matrix.
5. The next available caregiver is notified about the emergency.
6. The new caregiver acknowledges the alert.
7. The system records the escalation and acknowledgement.

4. Postconditions

1. The critical patient anomaly is recorded in the system.
2. An emergency alert is generated and stored.
3. The alert is sent to the appropriate caregiver.
4. If the first caregiver does not respond, the alert is escalated according to the caregiver matrix.
5. The alert status and acknowledgement details are recorded.